

Dawson Metzger-Fleetwood

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EDUCATION

University of Maryland

B.S., Computer Science with Machine Learning Specialization
B.S., Finance

College Park, MD

December 2022

SKILLS

Programming: Swift, Embedded Swift, SwiftUI, UIKit, RealityKit, TypeScript, React, HTML/CSS/JS, PHP, Python, C interop, SQL

Software: Xcode, VSCode, Node/Bun, Docker, Github Actions, TestFlight/Xcode Cloud, Jira, AWS (Lambda, DynamoDB, S3, CloudFront, Cognito, SES), Fly.io, HuggingFace, PyTorch, Claude Code, MCP

WORK EXPERIENCE

About Objects

Software Engineer

Reston, VA

July 2023 - Present

- Developed and maintained a multi-platform (iPad, iPhone, and Vision Pro) event management platform for government clients, featuring real-time resource tracking and 3D visualizations with RealityKit. Featured in Apple's enterprise showcase.
- Scaled the platform from inception to being able to handle 10,000+ concurrent users, architected real time data sharing for collaboration across distributed command teams, and optimized rendering hierarchical data with a tree based architecture that reduced latency by 80%.
- Scaled a CI/CD pipeline with GitHub Actions and automated TestFlight builds as the team quadrupled in size. Mentored junior engineers and interns.
- Led several internal projects, including the overhaul of the company website with custom WordPress plugins and a PHP backend, building an MCP server to help PMs with project administration, and building a pipeline to automate billing processes.
- Led development of a 3D ML object recognition pipeline, enabling rotation inference from 200 images. Supported an industrial oil drill manufacturer in analyzing drill bit component history and repairs.
- Led About Object's AI blog, writing articles on autoencoders, reinforcement learning, GNNs, and transformers.

Johns Hopkins University

Guest Lecturer and Consultant

Baltimore, MD

Spring 2024 - Present

- Designed accessible training materials and presented to interpreters on statistical concepts to enhance research interpretation, covering topics from fundamentals to the central limit theorem and multiple linear regression.
- Led development of applications designed to integrate internal systems and improve preparation efficiency, including a web app and MCP server for learning and rehearsing domain-specific vernacular ahead of technical talks, and a remote tooling server exposed over the internet via secure Cloudflare tunnels with OAuth 2.0.

Visual Language Associates, Inc

Technical Co-Founder

Silver Spring, MD

September 2019 - July 2023

- Co-founder of an agency that provides signed language interpreting and cued language transliteration services, holding contracts with Maryland and Virginia state governments, serving over two dozen clients in the DMV area with an average contract value of over \$90,000 and several million dollars in revenue.
- Led the automation and integration of client payment, payroll, and scheduling software to streamline operations.
- Responsible for coordinating scheduling with school officials and other interpreters/translitterators, as well as preparing and leading language and policy training sessions. Assisted with recruiting, onboarding, and sales.
- Developed methodology to screen employees, clients, and contracts, as well as match clients with service providers.

ADDITIONAL INFO

Languages: German (proficient), American Sign Language (proficient), Cued English (fluent)

Personal Projects

Silver Spring, MD

Gemma 4 MoE Fine-Tuning

Spring 2026

- Fine-tuned Google's Gemma 4 26B MoE model on an H100 using arbitrary rank ablation. Benchmarked two Pareto-optimal variants against Google's eval suite (MMLU, AIME 2026, GPQA Diamond, etc). Published on Huggingface.

Embedded Swift Agent

Fall 2025

- Built a CLI coding agent from scratch in Embedded Swift with C interop resulting in a 200 KB binary (macOS and Linux) and 120ms cold start. Supports parallel tool execution via pthreads, MCP client, subagents, web search, etc.

Helm

Fall 2025

- Published a VSCode/Cursor extension for workspace management, featuring Arc Browser inspired color coded spaces and swipe gestures, multi-root workspace support, and drag-and-drop reordering.

Startup Shell

College Park, MD

Fellow

September 2022 - Present

- Startup Shell is UMD's student-run startup incubator, founding home of ventures collectively worth over \$1 billion.
- As a fellow, assisted with product development for various ventures including an agentic engineering platform with 400K+ monthly visitors and 50K+ member Discord, a farmland valuation model, and clinical software for hospitals.

Boy Scouts of America

Colesville, MD

Eagle Scout

September 2012 - September 2016